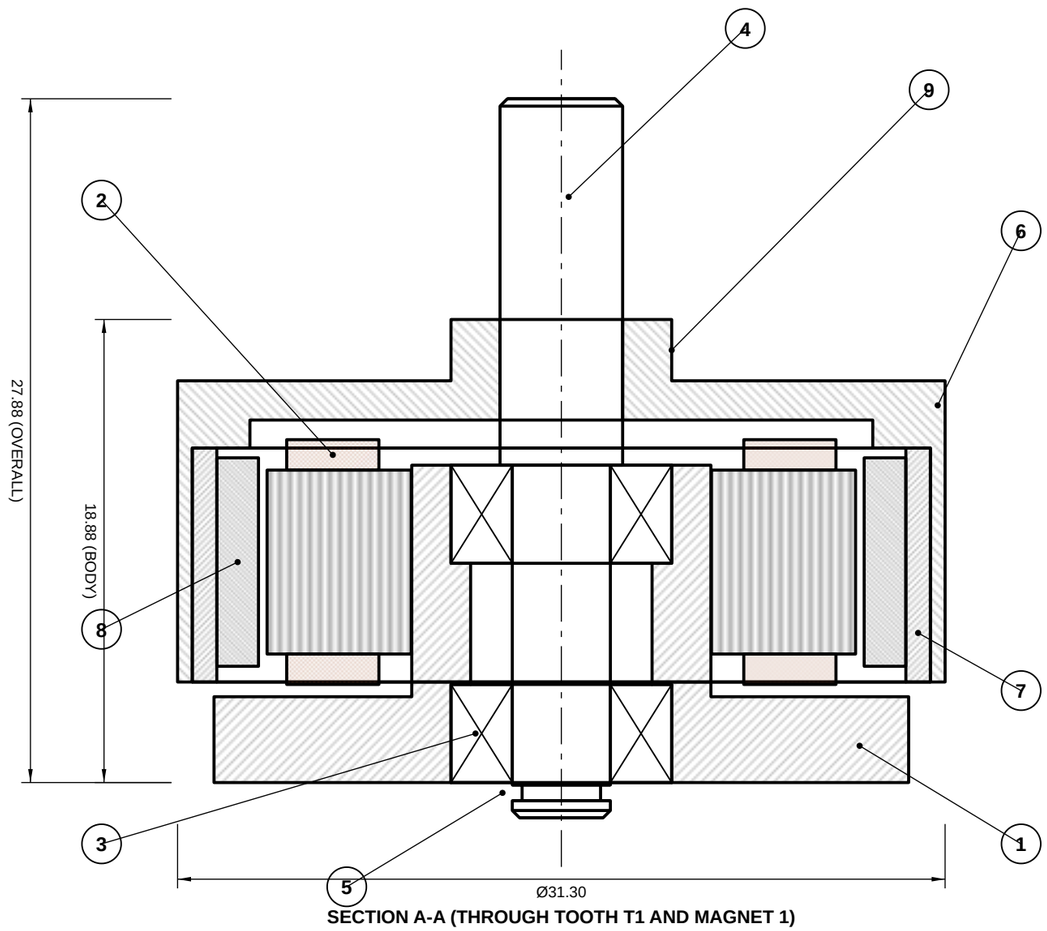
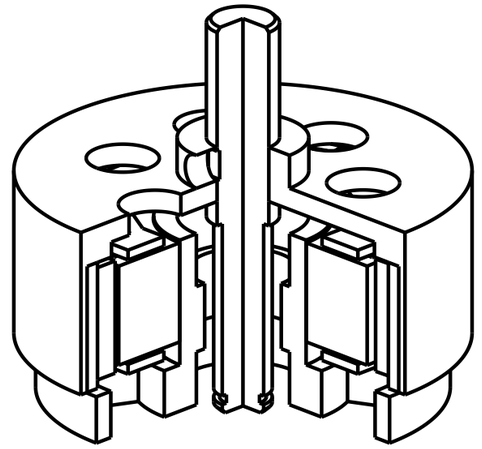


SECTION A-A (THROUGH TOOTH T1 AND MAGNET 1)



ISOMETRIC, QUARTER CUT-AWAY

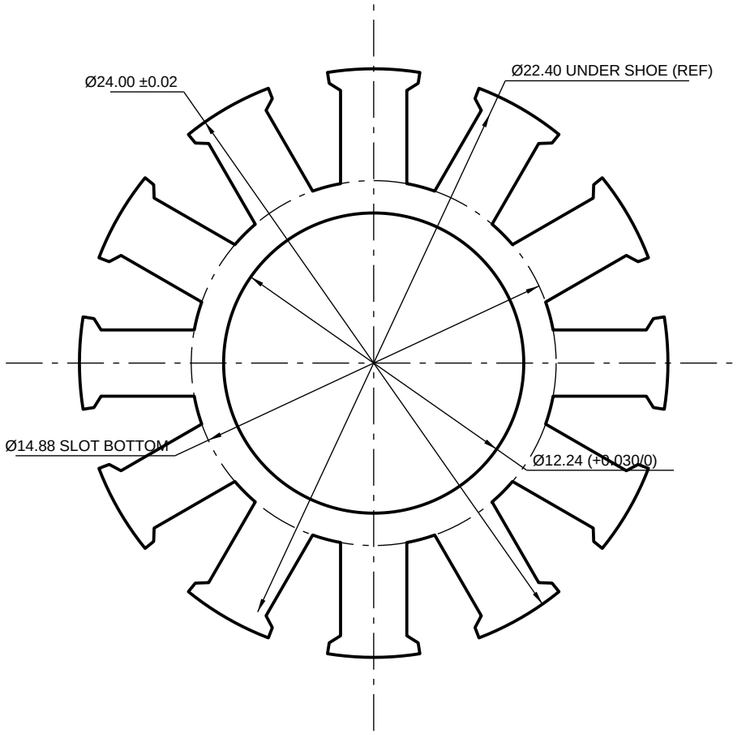
ITEM	PART NO	DESCRIPTION	QTY	MATERIAL / SPEC	SOURCE (CANDIDATES)
1	DWG-007	Base	1	7075-T6, anodized	CNC: Xometry / PCBWay / JLCCNC
2	DWG-003	Wound stator (incl. DWG-001/-002)	1	Si-steel + AWG23 Cu	Lams: Motorneo / Polaris / Precision Micro; wind in-house
3	684ZZ	Bearing 4x9x4 (NMB L-940ZZ / EZO)	2	ABEC-5+, shielded	EZO / NMB distributors
4	DWG-008	Shaft	1	52100, 58-62 HRC	Custom turn+grind (Misumi custom / CNC shop)
5	DIN 6799-3.2	E-clip	1	Spring steel	Fastener Superstore / McMaster
6	DWG-006	Bell	1	7075-T6, anodized	CNC: Xometry / PCBWay / JLCCNC
7	DWG-005	Back-iron ring	1	1010 steel	CNC/tube: same CNC vendor
8	DWG-004-01/-02	Arc magnet N-in / S-in	14	NdFeB, NiCuNi	K&J (proto) / Bestway / MPCO (volume)
9	ISO 4029 M3x3	Grub screw, cup point	1	Alloy steel 45H	Accu / McMaster
10	-	Shim washer 4x7x0.1	1	Spring steel	Misumi / McMaster
11	-	Lead wire 20 AWG silicone, 150 mm	3	200 C, 600 V	BNTECHGO / hobby suppliers

NOTES

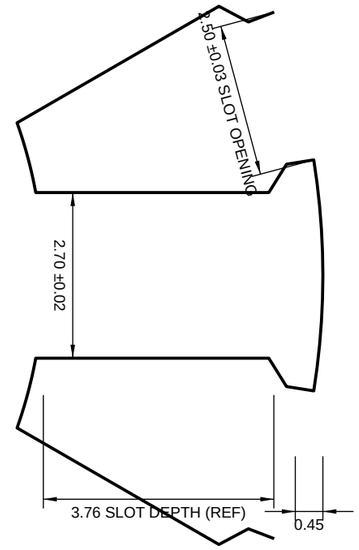
1. Press bearings (3) into base (1) with Loctite 641 on OD only; seat on shoulders with flat anvil, <= 300 N, no load through balls.
2. Bond stator (2) on tube with Loctite 638; align lead exit to window W1; cure 24 h RT or 15 min 120 C.
3. Bond magnets (8) into ring (7) with Loctite AA 326 + SF 7649 using alternating-pole indexing fixture; check pole order with compass.
4. Press ring into bell (6) (H7/s6, Loctite 638). Press shaft (4) into hub, fit grub (9) with Loctite 243 on flat.
5. Balance rotor (6+7+8+4): residual unbalance <= 11 mg·mm (ISO 21940-11 G2.5 at 36,000 rpm, rotor 16.6 g).
6. Insert rotor, add shim (10), fit e-clip (5). Axial play 0.02-0.08 mm; rotor turns freely, cogging uniform.
7. End-of-line test: R line-line, hipot, no-load Kv at 12 V (spec in spec_sheet.md), current, vibration, noise.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-000**
TITLE **MOTOR ASSEMBLY RL2407**	REV **P1**
MATERIAL **see parts list**	SCALE **4:1**
FINISH **-**	MASS **47.54 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **1 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



12 TEETH EQUALLY SPACED 30.0°



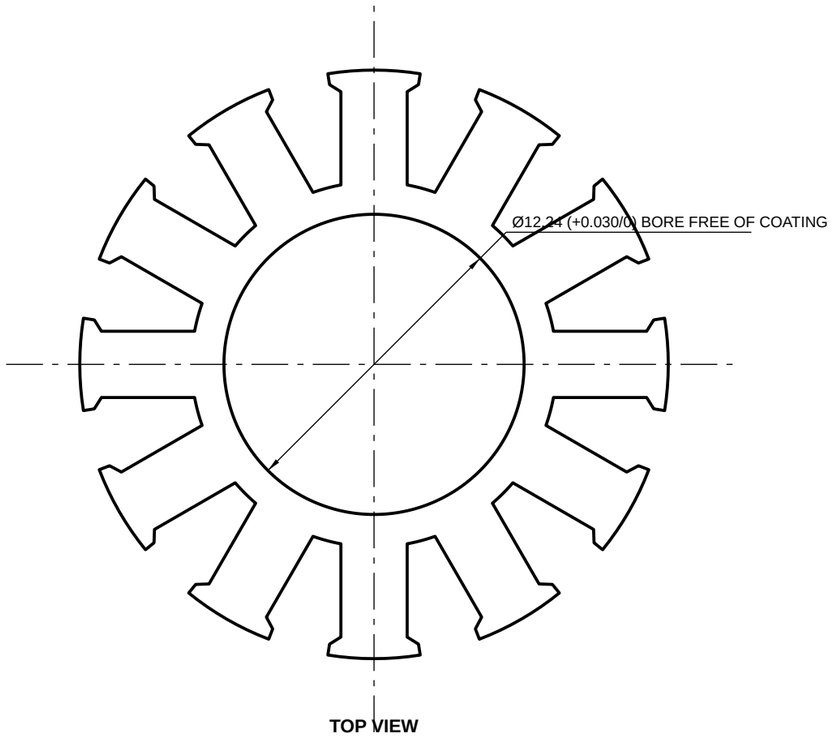
DETAIL A - TOOTH T1 (10:1)

NOTES

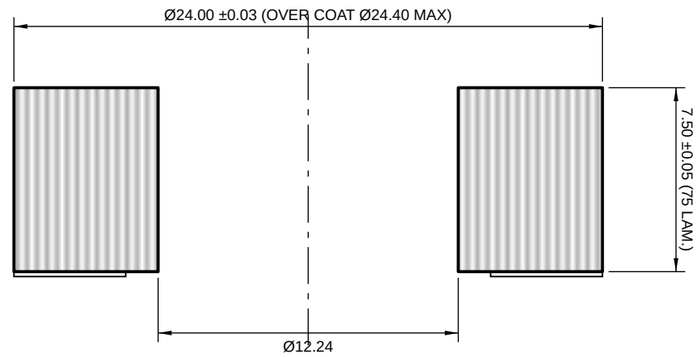
1. Material: 0.10 mm non-oriented electrical steel, 10JNEX900, NO10 or equivalent (P1.0/400 <= 13 W/kg).
2. Profile per lamination.dxf (1:1 mm). Profile tolerance 0.02 mm. Burr <= 0.01 mm. Flat within 0.02 mm.
3. Cut by wire-EDM, photo-etch or fibre laser; stress-relief anneal laser-cut parts (750-800 C, N2) unless vendor shows equal loss.
4. Quantity per motor: 75 pcs (7.50 mm stack). Keep grain/rolling direction marked, rotate alternate laminations 90 deg if anisotropic.
5. No sharp edges at slot bottom and shoe tips: R0.1 min to protect the magnet wire.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-001**
TITLE **STATOR LAMINATION**	REV **P1**
MATERIAL **0.10 mm NO Si-steel (10JNEX900 / NO10 or equiv.)**	SCALE **4:1, detail 10:1**
FINISH **C-5 inorganic insulation both sides, as rolled**	MASS **0.16 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **2 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



TOP VIEW



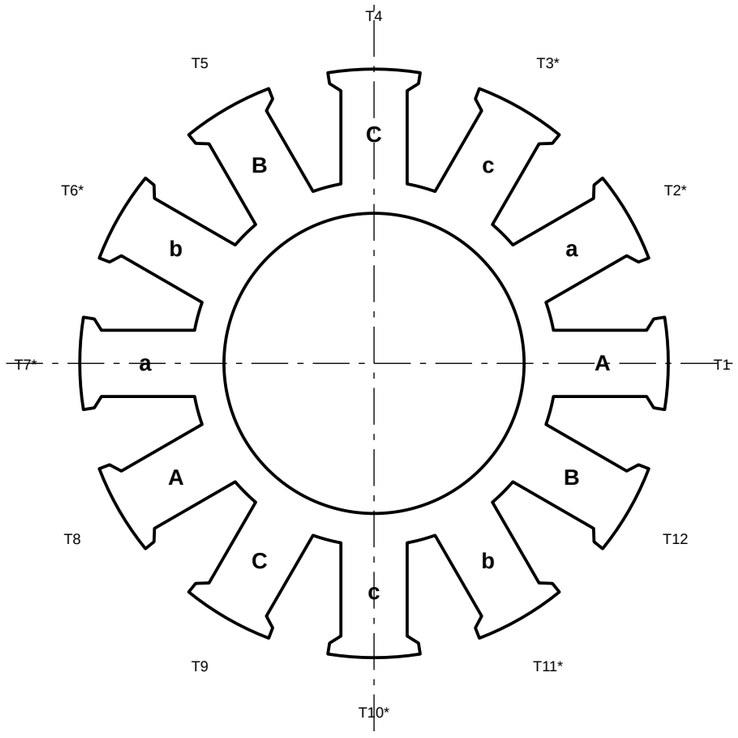
SECTION A-A

NOTES

1. Stack laminations on a mandrel; bond by self-bonding varnish (Backlack), glue-dot or 3 axial laser weld seams on the ID.
2. Stack height 7.50 ±0.05 mm under 0.5 MPa; teeth aligned within 0.03 mm; perpendicularity of faces to bore 0.03.
3. Coat all surfaces except bore: 0.15-0.25 mm, edges fully covered, no voids; dielectric >= 1.5 kV.
4. Bore Ø free of coating (mask or ream after cure) for bonded fit onto base tube (DWG-007).
5. Inspect: 100 % visual (edge coverage), 5 % section or spark test.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-002**
TITLE **STATOR CORE, BONDED + COATED**	REV **P1**
MATERIAL **75 x DWG-001**	SCALE **4:1**
FINISH **Epoxy powder coat (3M Scotchcast 265 or equiv.), class F min**	MASS **11.54 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **3 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



WINDING LAYOUT (VIEW FROM BELL SIDE)

WINDING TABLE

PHASE	COIL SEQUENCE (start -> finish)	TURNS
A	T1(CW) -> T2(CCW) -> T7(CCW) -> T8(CW)	6 / coil
B	T5(CW) -> T6(CCW) -> T11(CCW) -> T12(CW)	6 / coil
C	T3(CCW) -> T4(CW) -> T9(CW) -> T10(CCW)	6 / coil

ELECTRICAL

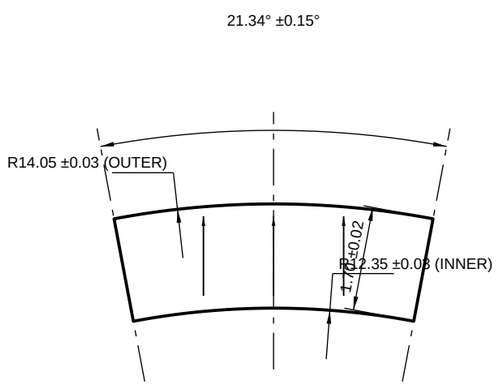
ITEM	REQUIREMENT
Connection	WYE (star point inside winding, sleeved)
Wire	AWG23 (0.573 mm bare / 0.609 mm max OD) heavy build, class 200
Turns	6 per tooth, 1 strand(s) in hand, layer-wound, no crossovers in slot
Slot fill (bare Cu)	0.38 (design); max 0.45
R line-line @25 C	87.7 mOhm ±6 % each pair; pair imbalance <= 2 %
Hipot winding-core	1.05 kV AC 1 s (2U+1000 V), leakage <= 1 mA
Insulation res.	>= 100 MOhm at 500 V DC
Leads	3 x 20 AWG silicone, 150 mm, tinned 3 mm, heat-shrink at joint

NOTES

1. Wind T1 first; '*' = reverse winding direction (CCW viewed from bell side). Inter-coil links routed on the base side.
2. Lead joints: strip enamel (solder pot 450 C or abrasive), solder, sleeve with PTFE/silicone tube; anchor with Kapton + varnish.
3. Impregnate: dip in Elmotherm 2340/70 (or equiv. class 180+) and cure per datasheet; no varnish in the bore.
4. Test 100 %: line-line resistance (4-wire, temperature corrected), hipot, IR, visual (no nicks, no wire above shoe).

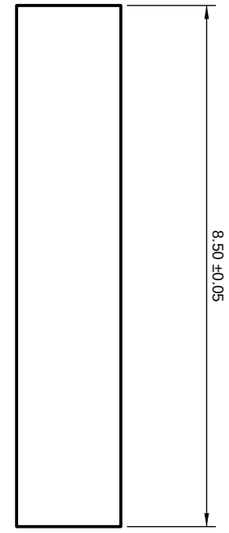
PROJECT **FPV OUTRUNNER**	DWG NO **DWG-003**
TITLE **WOUND STATOR**	REV **P1**
MATERIAL **DWG-002 + AWG23 magnet wire**	SCALE **4:1**
FINISH **Impregnated (varnish), class 180 min**	MASS **16.01 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **4 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



END VIEW

ARROWS = MAGNETIZATION DIRECTION FOR -01 (N ON CONCAVE FACE)



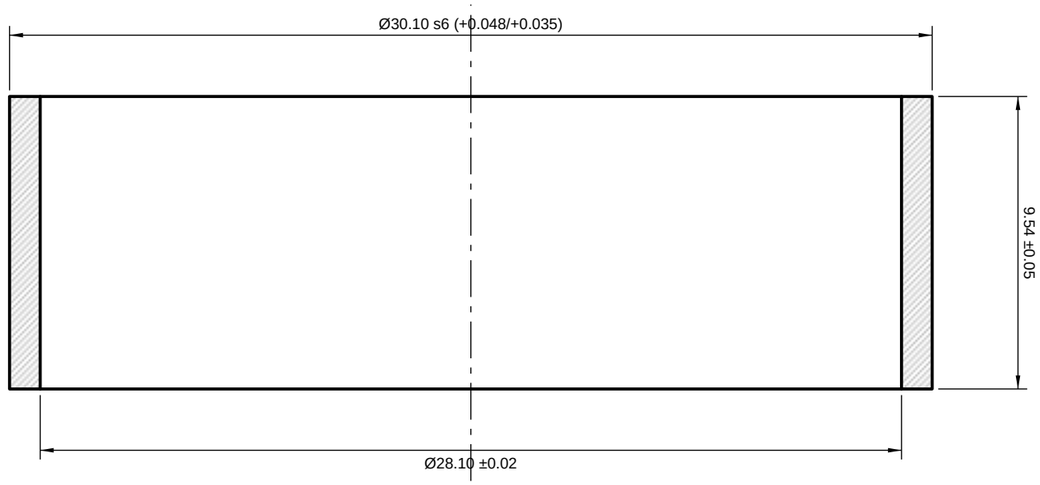
SIDE VIEW

NOTES

1. Grade N52: Br >= 1.42 T, Hcj >= 876 kA/m (N52). Max operating temp per grade; design burst magnet temp see design.json.
2. Magnetize radially. -01: N pole on concave (inner) face, marked with a dot on the outer face. -02: S pole on concave face, unmarked.
3. Supply as matched set: 7 x -01 + 7 x -02 per motor; flux of each piece within ±2 % of set mean.
4. Edges chamfered 0.1 x 45 deg max; no chips > 0.2 mm; coating free of blisters; salt spray 24 h.
5. Pack in alternating spacers; magnets are brittle and pinch hazards - handle with non-magnetic tools.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-004**
TITLE **ARC MAGNET (-01 N-IN / -02 S-IN)**	REV **P1**
MATERIAL **Sintered NdFeB N52 (alt. N48H/N45SH)**	SCALE **10:1**
FINISH **NiCuNi 10-20 µm**	MASS **0.53 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **5 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



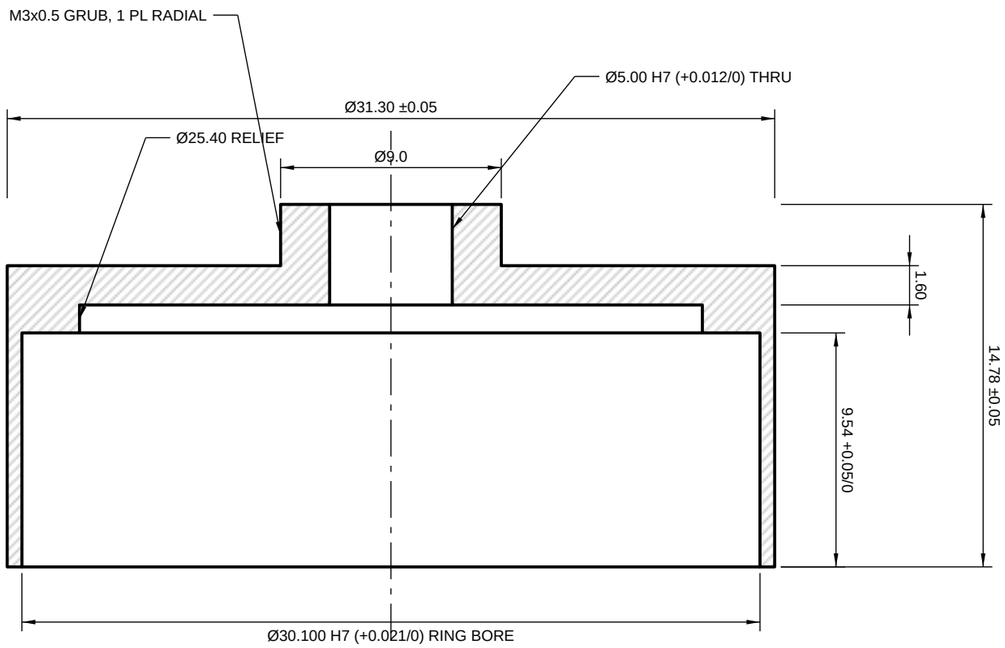
SECTION A-A

NOTES

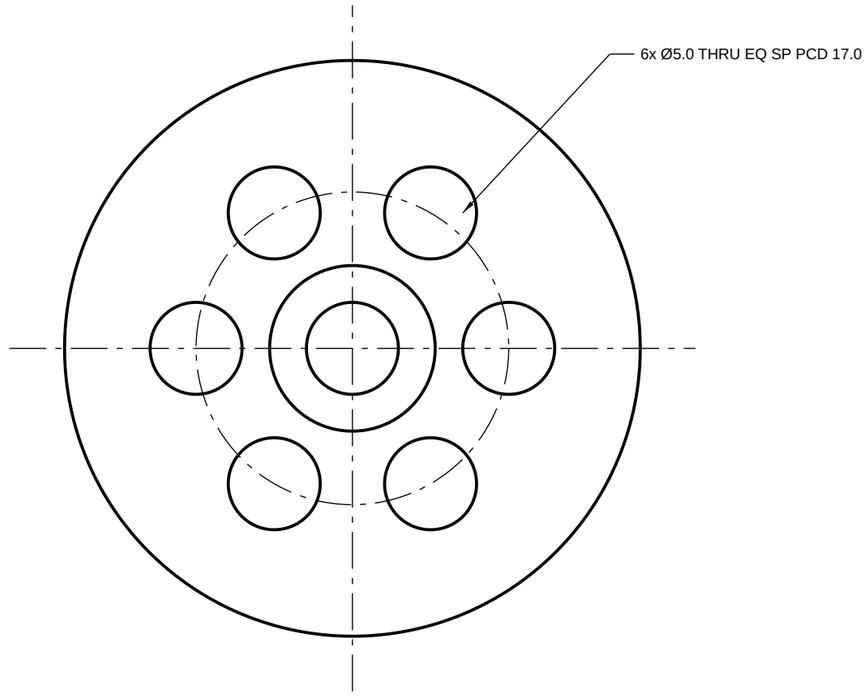
1. ID concentric to OD within 0.02 mm; faces perpendicular to axis within 0.03 mm.
2. Break edges 0.1-0.2 mm x 45 deg; ID edge on magnet-insertion side 0.3 x 45 deg lead-in.
3. Saturation: design flux density in ring per design.json (b_bell_t); do not substitute stainless (non-magnetic) steel.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-005**
TITLE **BACK-IRON RING**	REV **P1**
MATERIAL **Low-carbon steel 1010/1018 (or ST37), annealed**	SCALE **5:1**
FINISH **Zinc-nickel 5-8 µm or black oxide + oil**	MASS **6.86 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **6 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



SECTION A-A



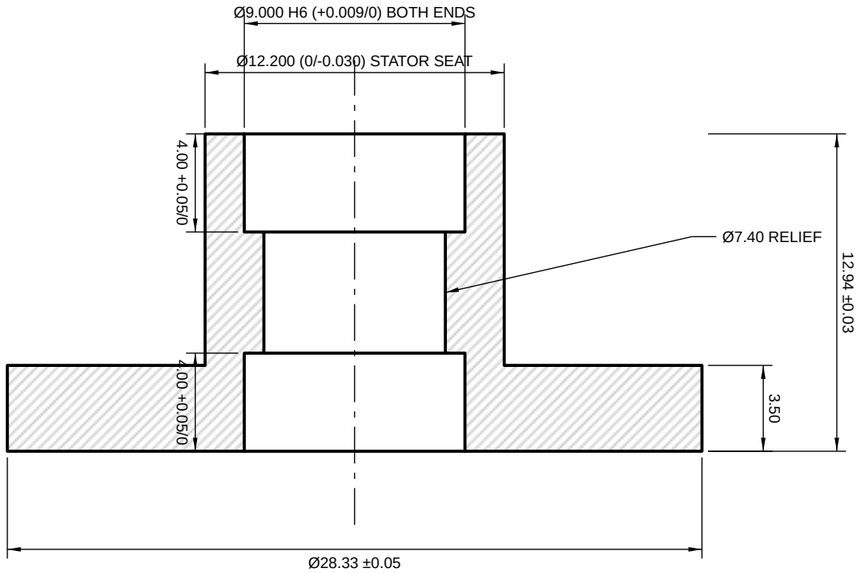
TOP VIEW

NOTES

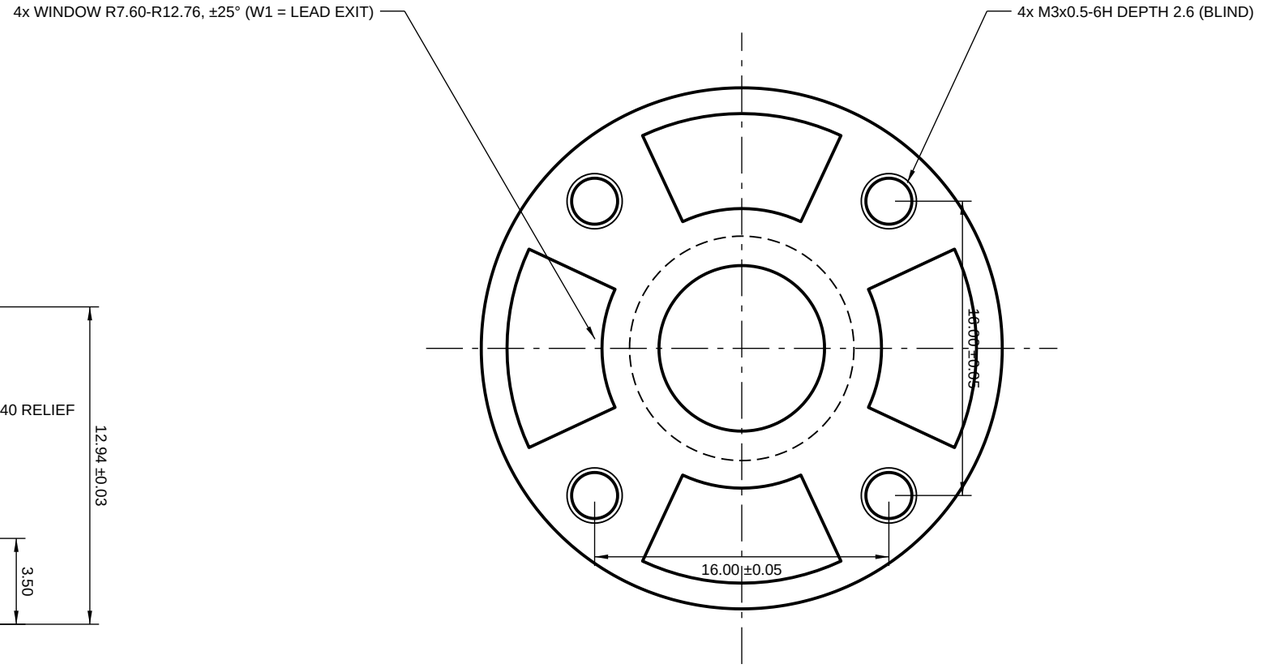
1. Ring bore and hub bore coaxial within 0.02 mm (A); plate top face perpendicular to hub bore within 0.02 mm.
2. Ring (DWG-005) is pressed in (H7/s6) with Loctite 638 or 648; anodize masked in ring bore.
3. Grub hole: tap M3x0.5 radially through hub boss on mid-height; deburr inside the bore.
4. Dynamic balance as assembled rotor, see DWG-000. Remove material only on skirt OD bottom chamfer zone.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-006**
TITLE **BELL**	REV **P1**
MATERIAL **Aluminium 7075-T6**	SCALE **4:1**
FINISH **Hard anodize type II, black, 10-15 µm (mask ring bore)**	MASS **5.52 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **7 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



SECTION A-A



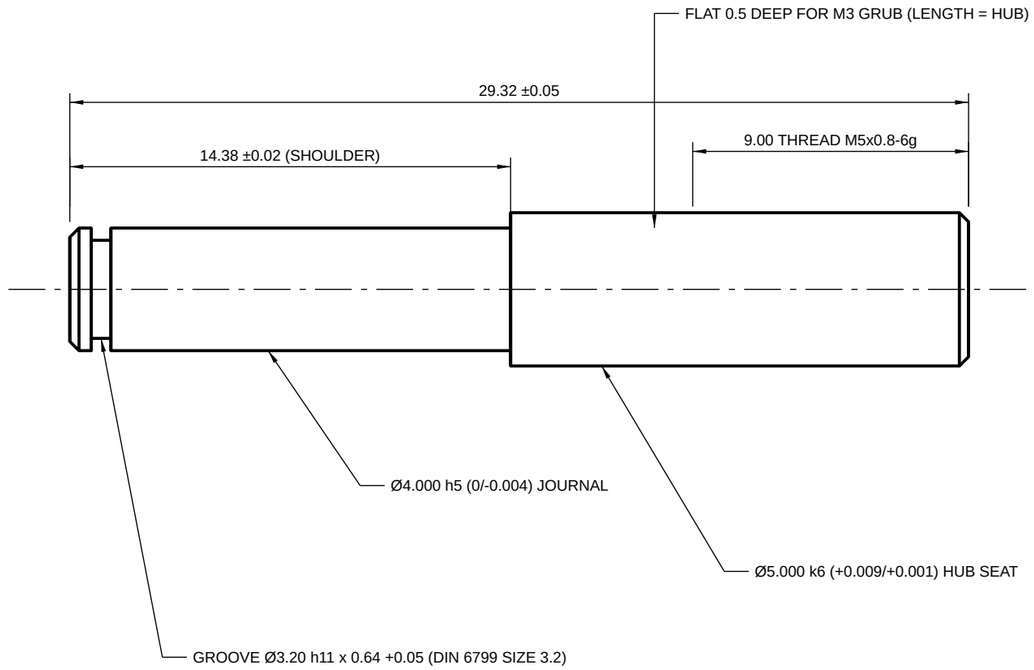
BOTTOM VIEW

NOTES

1. Bearing bores coaxial within 0.01 mm and to stator seat within 0.02 mm; machine both bores in one setup or line-bore.
2. Bearings 684ZZ retained with Loctite 641 (H6 housing). Stator bonded to seat with Loctite 638 (0.02-0.09 mm clearance).
3. Mounting threads blind; >= 0.8 mm floor protects end turns from over-length screws. Max screw engagement 2.6 mm.
4. Window edges break 0.2 mm; lead exit window W1 edges R0.3 min plus silicone grommet sleeve.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-007**
TITLE **BASE (STATOR MOUNT)**	REV **P1**
MATERIAL **Aluminium 7075-T6**	SCALE **4:1**
FINISH **Hard anodize type II, black (mask bearing bores and tube OD)**	MASS **5.32 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **8 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING



NOTES

1. Journal Ø4 h5 fits 684ZZ inner rings (push fit); runout of all diameters to journal axis 0.005 mm.
2. Shoulder face perpendicular to journal within 0.005 mm (it locates the upper inner race); undercut/relief R0.1 max.
3. Chamfer 0.3 x 45 deg both ends; thread M5x0.8-6g rolled or ground after hardening, no decarburisation.
4. Groove sharp-cornered (R0.05 max) for DIN 6799 e-clip; no burrs.

PROJECT **FPV OUTRUNNER**	DWG NO **DWG-008**
TITLE **SHAFT**	REV **P1**
MATERIAL **Bearing steel 52100 / SUJ2**	SCALE **5:1**
FINISH **Through hardened 58-62 HRC; journals ground Ra 0.4**	MASS **3.66 g**
TOLERANCE **ISO 2768-mK unless stated; dims mm**	SHEET **9 / 9**
DRAWN **bldc_motor.design.drawings 2026-11-05**	CHECKED **NOT CHECKED**

REV P1 - PROTOTYPE - GENERATED FROM PARAMETRIC MODEL - VERIFY AGAINST STEP BEFORE ORDERING